Chapter #4

The Future is Decentralized: Empowering Communities and Individuals

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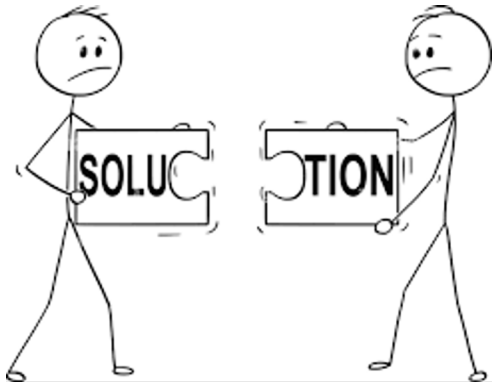
4.0 From Crisis to Innovation: The Cypherpunks and the Creation of a Decentralized Digital Currency

Before the creation of ***Bitcoin***, people were searching for ways to address the problems of traditional finance, such as fraud, corruption, and a lack of trust in financial institutions. These issues were made even more pressing by the global financial crisis of 2008. In response, a group of tech-savvy and forward-thinking individuals known as the Cypherpunks set out to create a **digital currency** that could be used for online transactions **without the need for intermediaries** like banks.

The Cypherpunks were rebels and visionaries who believed in the power of technology to bring about positive change and challenge traditional power structures. Many of them were involved in activism and civil liberties issues, and they were united by a shared passion for technology and a desire to use it to shape the future.



Q: How can individuals regain their financial self-sovereignty?

A: The Cypherpunks movement aims to create a new financial system that respects individuals' security, privacy, and freedom, as a solution to regain financial self-sovereignty.

And so, they set out to create ***bitcoin***, a digital currency that would revolutionize the way we think about money and financial transactions. To do this, they needed to find a way to record transactions that was more secure and transparent than traditional centralized ledger systems. Why did they feel this way?

4.1 Abuse of Centralization

4.1.1 Centralized Sytems

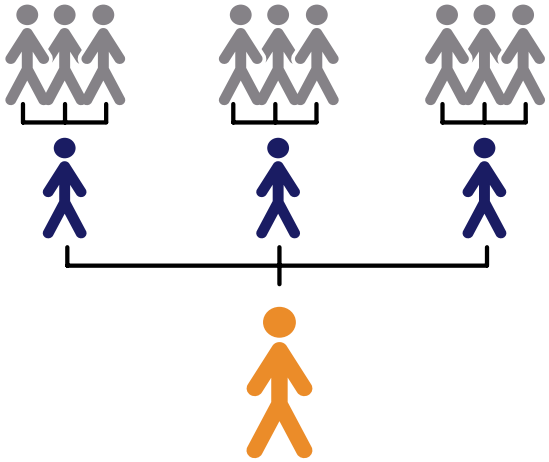
Centralization of power often leads to corruption, which can result in the mismanagement of resources, including financial resources. This can disproportionately affect those lower in the hierarchy and without as much influence or power, causing them to bear the greatest burden of the consequences of corruption and mismanagement.

The modern fiat system is characterized by centralization of control, with a small group of banks

and other financial institutions holding significant weight over the economy.

A centralized system can be thought of as a tree with a single trunk. The trunk represents the central authority or point of control, and the branches represent the various parts of the system that are controlled by the central authority. In this analogy, the tree is vulnerable if the trunk is damaged or diseased, because the entire tree relies on the trunk for support.

There are many **drawbacks to centralized systems** including:



- **Vulnerability**: A centralized system relies on a single point, so if that point fails, the whole system can fail.
- **Control and power**: Those in control of centralized systems have a lot of power and influence over how they work.
- **Inefficiency and intermediaries**: Centralized systems often use intermediaries, which can make them slow and add extra costs.
- **Lack of autonomy**: People may not be able to make their own financial decisions.
- **Censorship and restriction**: There is a risk of being blocked or restricted from accessing certain financial resources in centralized systems.
- **Scaling challenges**: Centralized systems may struggle to keep up with increasing demand for financial services and resources.
- **Security risks**: Centralized systems can have weaknesses that hackers can use to gain access or cause damage.
- **Lack of transparency and trust**: It can be hard to understand how centralized systems work and to make informed decisions about them because they may *not be transparent or trustworthy*.

4.1.2 Counting the Middlemen: A Look at the Intermediaries in a Credit Card Transaction

Modern banking. Simple, right? Take something as seemingly simple as buying a hamburger with a credit card. At first glance, it may seem undemanding and harmless. But if we break down the steps and see the intermediaries involved, you might be surprised at what we discover. Are there inconveniences, inefficiencies, maybe even hidden dangers lurking in the shadows? Let's find out.

Step	Transactions	Descriptions
1	Cardholder-Merchant	You go to McDonald's and order a hamburger using your Citi MasterCard card.
2	Merchant-Payment Processor	McDonald's sends an authorization request to its payment processor.
3	Payment Processor - Credit Card Network	The processor receives the request and sends it to Mastercard.
4	Credit Card Network - Issuing Bank	Mastercard passes the request along to your issuing bank, CitiBank.
5	Issuing Bank - Credit Card Network	Citibank checks that your account is in good standing and sends the authorization code back to the Mastercard.
6	Credit Card Network - Payment Processor	Mastercard sends the authorization back to the processor.
7	Payment Processor - Merchant	The processor sends the authorization back to McDonald's.
8	Cardholder - Merchant	You receive your hamburger.

1

Cardholder
Presents a credit card to the merchant as payment.

2

Merchant
The merchant sends the transaction details to its payment processor.

3

Payment Processor
The processor relays the transaction data to the card network.

4

Credit Card Network
Sends an authorization request to the issuing bank.

5

Issuing Bank
Verifies the card details, checks for available funds, and sends its response (approved or declined) to the cade network.

6

Credit Card Network
Sends the issuer's response to the merchant's payment processor.

7

Payment Processor
Relays the issuer's response to the merchant.

8

Merchant
The merchant and the cardholder complete the transaction.

At this point, no actual funds have changed hands, except perhaps a small authorization fee. The transaction only exists on "paper." McDonald's needs to close or batch-out its sales for the day. The closing process might look like this:

1. McDonald's terminal or point-of-sale (POS) system sends the day's transactions to the processor.

2. The **processor** sends the transaction information to **Mastercard**.

3. **Mastercard** sends the transactions to **Citibank**.

4. **Citibank** confirms the authorizations, **holds back their interchange fees** (there are over 900 possible fee codes in North America), and transfers the funds back to **Mastercard**.

5. **Mastercard takes its assessment fee** and sends the funds to the **processor**.

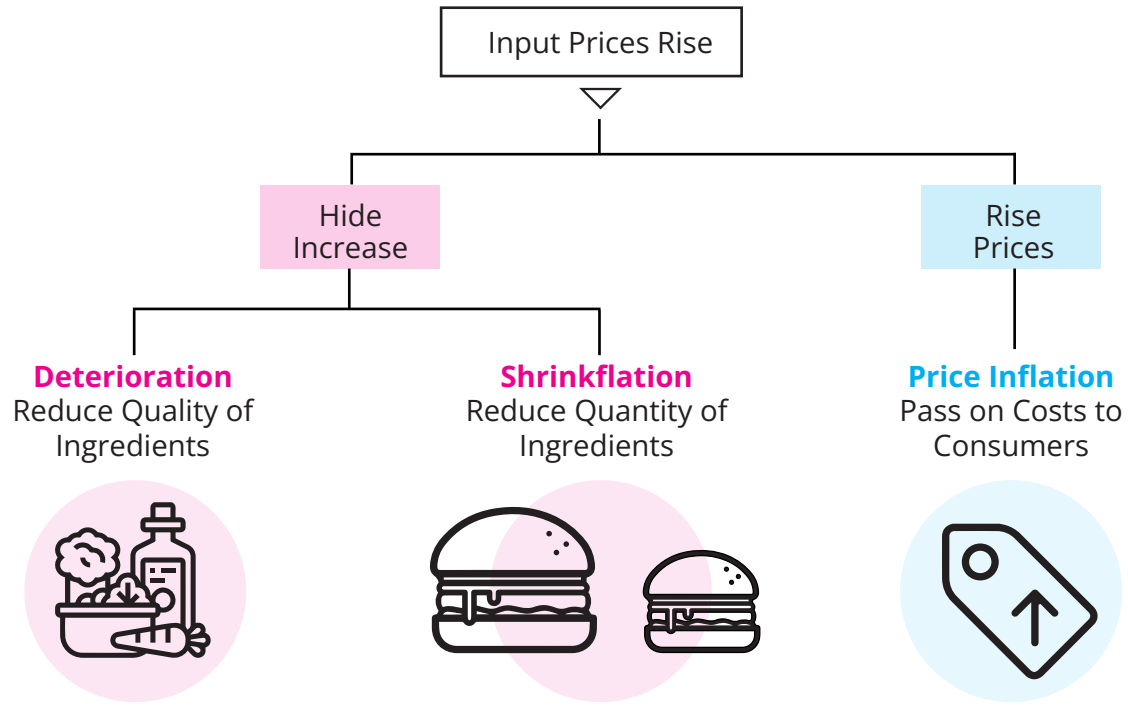
6. The **processor takes its cut**, as set out in the merchant agreement, and deposits the funds to McDonald's bank account.

Who do you think paid for the fees? Of course YOU. But, did anybody inform you of this? Oh, no! They were hidden in the cost of the hamburger.

And all of this happens believe it or not because we rely on centralization.

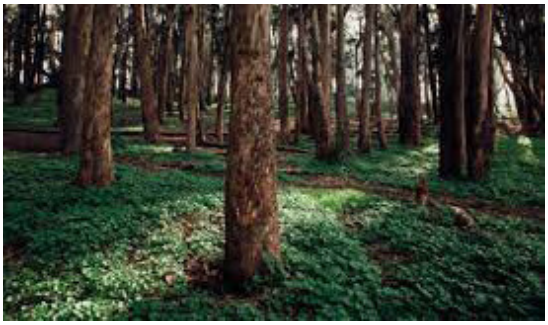
The modern banking world comes with various risks, including accidental double swipes, credit card fraud, human and computer errors, and potential hacks.

Disguising Inflation



4.2 A Powerful Tool for Overcoming the Limitations of Centralization

Decentralized systems, on the other hand, can be thought of as a forest. Each tree represents an independent participant, and the forest represents the overall system. In this analogy, the forest is more resilient than an individual tree because it does not rely on a single point of failure. If one tree is damaged or diseased, the rest of the forest can continue to thrive. The trees in the forest share the ground, the nutrients, the sun, and the rain.



Decentralized systems, like communities, networks, and forests, function best when there is a diverse group of individuals working together, rather than a single central authority dictating all the rules.

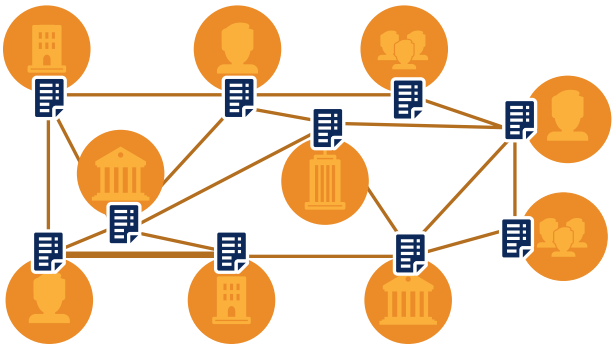
A **network** is a group of **nodes** that are connected to each other in some way. This connection allows the devices to exchange information and communicate with each other.

A **node** is a *computer connected to a network* that can *share and/or receive information* and communicate with the other nodes.

Advantages of a decentralized system:

- It is more resilient and reliable because there is ***no single point of failure***. If one part of the system goes down, the rest can continue to operate.
- With the right encryption, decentralization is more ***secure*** because there is no central point of control that can be targeted by hackers.

Distributed Ledger Technology



- It can help you become ***sovereign***, meaning that you'll have more control and autonomy over your own assets and decisions, rather than relying on a central authority
- They can be more ***transparent*** because all nodes have access to the same information and can see what is happening in the system.
- They can be ***permissionless***, meaning that anyone can join or participate in the system without needing permission from a central authority.
- They can be ***limitless***, meaning that there is no predetermined limit on the number of nodes that can join the system.
- Each node has ***equal opportunities*** to contribute and influence the network, making it a more democratic and inclusive structure.
- Participants can also choose to use pseudonyms or "nicknames" to ***protect their privacy and security***, which can make the system more resistant to censorship and attacks.

Decentralized scarcity is often seen as a good thing for money because it prevents inflation and manipulation by a central authority.

However, decentralized systems also have their **challenges** and limitations.

- Decentralized systems may require more work to get all the connected devices (nodes) to agree and work together.
- Decentralized systems may also be more at risk for trouble caused by bad actors or devices (malicious nodes) that could harm the network.

4.2.1 Class Exercise: Decentralized Consensus Game with Bad Actors

In a decentralized network, **consensus** refers to the process of reaching agreement among the members of the network. This can present difficulties as there is no central authority to make decisions or resolve conflicts. Instead, decisions must be made through a process of negotiation and compromise among the members of the organization.

Class Exercise. In this game, you will be playing the role of **nodes** in a decentralized network. Your goal is to come to a **consensus** on a problem **without trusting each other**.

- You will play the role of a node in a decentralized network and work with others to come to a consensus on a problem.
- There may be bad actors in the group who will try to mislead or sabotage the process.
- As a **good actor**, your goal is to work with others to verify information and achieve consensus.

- As a **bad actor**, your goal is to mislead the group but do so in a subtle manner.
- The purpose of the game is to understand the challenges and benefits of decentralized systems and learn how to verify information, achieve consensus, and identify malicious behavior.
- You will be divided into small groups and given a problem to solve within a set time frame.

Remember, **in a decentralized system, you cannot simply trust the answers of other group members**. You must verify the accuracy of the information and come to a consensus through discussion and collaboration.

4.3 Transactions are Just Agreements to Trade

Welcome to *The **Decentralized** Micronesian island of Yap!* It is a bit remote, but fascinating because people use a special kind of currency called "Rai stones." A feature that makes them a great form of money is their **scarcity**. The total number of Rai stones is *limited*, which means that *they cannot be easily reproduced or inflated* like fiat currencies. This fixed supply helps to maintain the purchasing power of the Rai stones over time and makes them a reliable store of value. These Rai stones are like giant coins that are used to purchase stuff on the island. The thing is, they can weigh a *ton*. Rai stones can actually crush you, so they're a bit impractical to carry around. How then, can people conveniently use Rai stones as mediums of exchange without having to physically take them from one place to another?

4.3.1 To Trust or Not to Trust

While the US Dollar is now the official currency of Yap Island, Rai stones still are a type of money. Unlike the dollars, the Rai stones on Yap Island are not controlled by a single authority or stored in banks. Instead, the transactions are based on oral history and trust, with people keeping track of their own records of who owns which stones.

This system has both benefits and drawbacks. On the one hand, it allows for a certain degree of independence from one central authority. On the other hand, it can also lead to disagreements and potential for cheating. Why?

Decentralization is easy to achieve in small groups. Life is simple as there are fewer people to coordinate; it is often possible for everyone to have a say in decision-making processes and for those

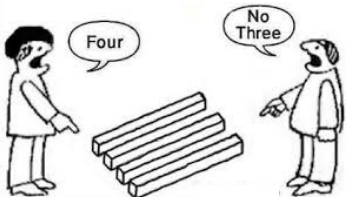
decisions to be implemented relatively quickly. As a group gets larger, it becomes more difficult to reach an agreement and for decisions to be implemented effectively.

• Imagine that you have a field full of ripe corn that needs to be harvested. You need some help, so you approach your neighbor, Raquel, and offer her a deal: if she helps you harvest the corn, you will give her a 10 kg stone in exchange. Raquel agrees, and for the next day, she works alongside you in the fields, helping to gather the corn and bring it in. At the end of the day, you both shake hands and instead of physically handing over the stone, you simply show her that her payment (the Rai stone) is in your backyard.

• From that point on, *you both agree* that the stone now belongs to Raquel. This type of **transaction**, where no currency actually is handed from one individual to the other as a form of payment, but instead a *physical object is used as a* ***symbol of value***, is common on Yap Island and has been used for centuries as a form of currency.

• Five years later, you decide to try and claim the Rai stone as your own. You present evidence to the community that the stone has been passed down through your family for generations, and that you are the rightful owner.

• However, Raquel remembers the agreement that you both made and provides evidence by bringing witnesses of the exchange to give a statement. She argues that the stone rightfully belongs to her, as it was given to her in exchange for her help with the harvest.



• Some members of the community might agree with your claim, citing the tradition and history of your family's ownership of the stone. However, others might side with Raquel, pointing to the agreement that was made and the fact that the stone has been in her possession (figuratively speaking) for five years without any objections from other members of the community. Factors that might be considered include the history and tradition of ownership, the terms of the agreement between you and Raquel, and any relevant evidence or arguments. Not a very solid solution, is it?

So then, how can thousands of strangers all agree on one truth without anyone having the final say? This is something that has puzzled people for a long time, and it's an important question to consider. It turns out that the internet has helped us find a solution to this problem. The solution is called the **blockchain**.

4.3.2 Let's Swap Trust for Rules

Imagine you and your friends are in a group chat where you can buy and sell things with each other. Every time a purchase is made, it's recorded in a shared document for everyone to see and each person's balance is updated. This chat uses a digital ledger to keep track of all the transactions that have happened. The ledger is like a record book that everyone can see.

In a decentralized system like this, all the participants have a copy of the ledger. This makes it hard for any one person or group to change any information without being noticed. It's like a security measure to make sure that the records are accurate and no one can cheat. This is similar to how a **blockchain** works.

Instead of relying on personal relationships and subjective interpretations of trust, a decentralized system can operate effectively if it is *based on a set of clear, transparent rules that everyone agrees to follow*. This way, *decisions can be made and conflicts can be resolved in a fair and objective manner, without relying on the trust of individual parties*. It might not be as romantic as relying on trust, but it's a much more reliable way to ensure that a decentralized system operates smoothly.

- If Yap Island had a set of unbreakable rules and a written record of all transactions between its members, the conflict between you and Raquel could have been avoided. These rules and records would have made it clear to all members of the village what their rights and responsibilities were.

But is it that simple? Not really; there was a lot of trial and error before blockchain technology was actually a success.

- What are the exact rules that must be followed?
- Who makes these rules?
- Why will people want to follow the rules?
- How do rules get distributed across the network?
- What will happen if someone breaks the rules?
- How can the rules be changed or updated later?
- How will the rules be enforced to make sure everyone follows them?
- How can the rules be made clear and easy to find for everyone in the system?

4.4 Unlocking the Power of the Blockchain: A Technology Revolutionizing the Future

Despite numerous setbacks, one very enigmatic person (or group of people) finally found the key to developing a game-changer methodology for the world of trade and finance. This masterpiece made it incredibly easy to track and verify transactions, streamlining the process of exchanging money, goods, and other assets. With its innovative approach and advanced technology, this system revolutionized the way we think about economic transactions, making them faster, safer, and more efficient than ever before.

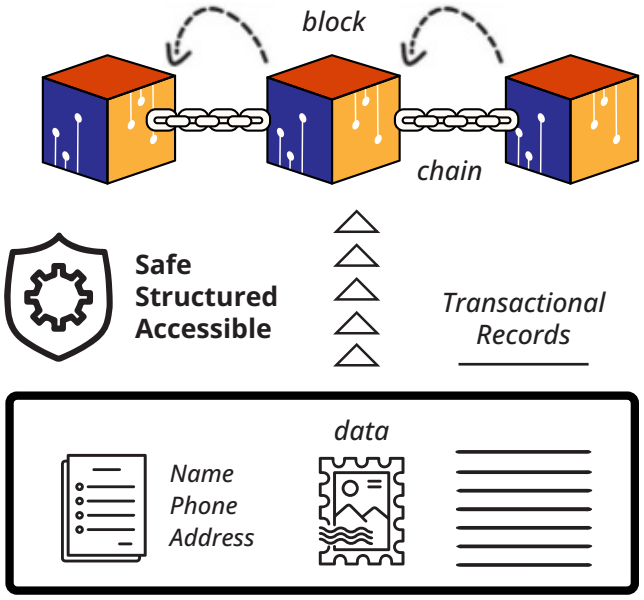
A ***blockchain*** is a decentralized digital ledger that securely records and verifies all transactions across multiple computers in a transparent manner.

A **blockchain** is like a history book. Each page (or "**block**") has a list of things that happened (**transactions**). As more things happen, we need to add new pages (blocks) to the book. Anyone can read the book for free, but only special helpers (**miners**) can add new pages. They make sure that what's written is true. Once something is written in the book, it can't be changed or erased. It's a permanent record of all the **transactions** that have happened on the **blockchain**.

- A **blockchain** does not have a central authority (like an author, a publisher or an editor) that can edit, delete or change the information recorded in it, hence it's considered a more secure and reliable method of record keeping compared to a traditional central database.

What is a blockchain?

All records of actions on the blockchain are called transactions.



If the helpers (**miners**) are not in **consensus** about the validity of the pages (**blocks**), they will be rejected and will not be added to the **blockchain**.

But in order to understand the **blockchain**, we have to get a sense of the context in which it exists. While many feel that the **blockchain** has uses as an independent innovation, its true founding role is singular: to create an immutable ledger so that a decentralized, trustless form of money exists. To understand the **blockchain**, we need to understand ***Bitcoin*** as a whole.